SC2006 Lab 3 — AI-based Project Initialisation

Team 5 · GoWhere · ChatGPT (free tier) · 29 October 2026

Part 1 — Technology stack re-prompt (3.3.1)

Uploaded: our current requirements document and the current class diagram. In Lab 2 we uploaded only the Lab 1 deliverable: the one-feature Neighbourhood Comparison App, with no class diagram.

Prompt (same as Lab 2, in a fresh chat):

> Based on the uploaded design materials, recommend a technology stack appropriate for an entry-level software engineering team

AI response:

Recommended Technology Stack for GoWhere

1. Executive recommendation

Based on the GoWhere requirements document and class diagram, I recommend a **Python-based monolithic web application with server-rendered HTML, lightweight JavaScript, SQLite, and automated testing using pytest**.

For an entry-level software engineering team, the goal should be to choose technologies that are easy to learn, straightforward to maintain, and capable of meeting GoWhere's actual requirements without introducing unnecessary infrastructure.

GoWhere is a web application with three interconnected features:

- **Where to Live:** Compare 2–4 Singapore neighbourhoods using six weighted factors and recommend the best match.
- **Where to Lepak:** Find upcoming events based on categories, dates, travel time, and transport mode.
- **Where to Eat:** Find food establishments based on location, cuisine, opening hours, and travel preferences.

The application also integrates external services, including OneMap, LTA DataMall, Telegram, Google Places, and an LLM service. Its requirements emphasise reliability when external services fail, modularity, data accuracy, privacy, and response time. [cites: GoWhere_requirements_and_use_ca…]

My central recommendation is to keep the application as a modular monolith rather than split it into microservices. The class diagram already separates boundary adapters, control logic, and entities, providing a useful foundation for a small team to develop and maintain the system. [cites: class_diagram_mermaid]

2. Recommended stack at a glance

Layer	Recommended technology	Why it fits GoWhere
Programming language	Python	Suitable for application logic, data processing, scoring algorithms, and API integration.
Backend framework	Flask	Lightweight and approachable for a small team building a server-rendered web application.
Frontend	HTML5, CSS3, vanilla JavaScript	Supports interactive forms, filtering, maps, and charts without a complex frontend build system.
HTML templating	Jinja	Allows reusable page layouts and dynamic rendering of search results.
Data analysis	pandas, NumPy	Useful for neighbourhood statistics, weighted calculations, and data preparation.
Primary database	SQLite	Simple deployment and maintenance for a modest application with structured data.
Data visualisation	Chart.js	Suitable for the grouped bar charts required by the neighbourhood comparison feature.
Mapping	OneMap API	Provides Singapore-focused maps, location resolution, and route information already required by the design.
External API integration	Python `requests`	Keeps HTTP requests to external services simple and consistent.
Automated testing	pytest	Supports unit, integration, and regression testing of business logic and external-service adapters.
Version control	Git and GitHub	Supports collaboration, code review, issue tracking, and change history.
Code quality	Ruff and a formatter such as Black	Helps maintain consistent, readable Python code.
Dependency management	Python virtual environments and a pinned requirements file	Helps teammates reproduce the same development environment.
Deployment	A managed Python hosting service or a small Linux server	Avoids unnecessary infrastructure and operational complexity.
Automation	A scheduled job or cron	Supports periodic event ingestion and data-maintenance tasks.

This is a proposed stack, not a claim that every listed technology is already used by the implementation. The supplied design documents describe the system's behaviour and architecture but do not establish every technology choice in this table.

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3. Backend: Python and Flask

Recommendation: Python + Flask

Python is a particularly suitable choice because GoWhere combines conventional web development with data processing, scoring, geographic calculations, and LLM integration.

The class diagram already identifies separate components for:

- Web request handling (`WebRoutes`)
- Search logic (`LepakSearch` and `EatSearch`)
- Neighbourhood scoring (`ScoringEngine` and `ScoringStrategy`)
- Location and routing (`LocationService`, `EventRouter`, and `CommuteRouter`)
- External services (`OneMapAdapter`, `DataMallAdapter`, and `LlmAdapter`)
- Data access (`Snapshot`, `EventStore`, and `PlaceStore`)

These responsibilities can be organised into Flask routes, Python service modules, data-access modules, and external-service adapters without requiring a large framework. [cites: class_diagram_mermaid, class_diagram_mermaid]

Why Flask rather than a more complex framework?

For an entry-level team, Flask offers several advantages:

1. **A small learning curve:** Developers can understand request handling, routing, and templates without learning a large framework first.
2. **Flexible architecture:** The team can retain the existing separation between boundary, control, and entity components.
3. **Straightforward integration:** Calling external APIs and processing their responses fits naturally into Python modules.
4. **Incremental development:** Each feature can be developed and tested separately while sharing common services.

Django would also be a reasonable choice, particularly if the application needed built-in authentication, an administrative interface, and a more extensive data-management layer. However, GoWhere explicitly does not require account creation, and its current design already has a relatively focused set of application responsibilities.

I would choose Flask unless the team has a specific requirement that Flask cannot meet conveniently.

Suggested project structure

```
gowhere/
├── app/
│   ├── routes/
│   │   ├── live.py
│   │   ├── lepak.py
│   │   └── eat.py
│   ├── services/
│   │   ├── scoring.py
│   │   ├── location.py
│   │   ├── routing.py
│   │   ├── parking.py
│   │   └── ingestion.py
│   ├── adapters/
│   │   ├── onemap.py
│   │   ├── datamall.py
│   │   ├── telegram.py
│   │   ├── google_places.py
│   │   └── llm.py
│   ├── repositories/
│   │   ├── events.py
│   │   └── neighbourhoods.py
│   ├── templates/
│   ├── static/
│   │   ├── css/
│   │   └── js/
│   └── __init__.py
├── data/
├── tests/
├── scripts/
├── requirements.txt
├── .env.example
└── README.md
```

This is a suggested organisation rather than a requirement to reproduce every existing class as a separate file. The important principle is that **routes should handle web requests, services should implement business rules, repositories should manage data access, and adapters should communicate with external services**.

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4. Frontend: HTML, CSS, JavaScript, and Jinja

Recommendation: Server-rendered pages with lightweight JavaScript

I would start with:

- **HTML** for page structure and forms.
- **CSS** for layout, responsive design, and consistent styling.
- **Vanilla JavaScript** for interactive controls and asynchronous updates.
- **Jinja** for rendering Python data into HTML.
- **Chart.js** for the neighbourhood comparison chart.

I would not introduce React, Next.js, or a separate frontend application at the beginning.

Why this approach fits GoWhere

GoWhere has three distinct features, but most of their interactions are form-driven:

- Selecting neighbourhoods and configuring comparison weights.
- Selecting event categories, dates, travel modes, and maximum travel times.

- Filtering food establishments and sorting by distance, opening status, or rating.

These workflows can be implemented with server-rendered pages and targeted JavaScript enhancements.

This approach also supports the requirement that the application remain usable when certain presentation features are unavailable. For example, the requirements explicitly state that neighbourhood selection must remain usable when the map cannot be displayed. [cites: GoWhere_requirements_and_use_ca…]

A practical implementation would use:

Feature	Implementation
Search forms	HTML forms with server-side validation
Dynamic filters	JavaScript
Search progress indicators	JavaScript, with progress responses or asynchronous requests from the backend
Neighbourhood ranking	Server-rendered HTML
Grouped bar chart	Chart.js
Neighbourhood boundaries	OneMap-compatible mapping implementation using locally held boundary geometry
Event and food routes	OneMap API, with the resulting route details rendered in the interface
Responsive layouts	CSS media queries and flexible layouts

When would React become worthwhile?

Consider React if the application later requires extensive client-side state management, highly interactive map interfaces, or a much richer single-page experience.

For the current project, adding React would introduce another language ecosystem, a build process, frontend-backend integration concerns, and additional testing requirements.

Those costs are difficult to justify before the team has demonstrated that a simpler interface is insufficient.

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5. Database and data management

Recommendation: SQLite initially

SQLite is a reasonable starting point because GoWhere maintains structured neighbourhood, event, and place data, while the class diagram already separates those data responsibilities.

The requirements also distinguish between relatively stable datasets and information that changes frequently. For example, neighbourhood data is refreshed through a dedicated procedure, while event ingestion is performed periodically and parking availability must be no more than two minutes old. [cites: GoWhere_requirements_and_use_ca…, GoWhere_requirements_and_use_ca…]

I would organise the data into three main groups.

Data group	Suggested storage	Reason

Neighbourhood snapshot	SQLite database	Structured data used by scoring and comparison logic
Event records	SQLite database	Supports event lookup, updates, deduplication, and expiry
Place list	SQLite database or a validated local dataset	Supports cuisine filtering, location searches, and place details
Session state	In-memory session store initially	Supports temporary user preferences and search state
API credentials	Environment variables or hosting-platform secrets	Keeps credentials outside the source code
Live parking data	Short-lived cache	Meets the two-minute freshness requirement
Place ratings	Retrieved on demand, not persisted	Respects the requirement not to store rating data

A key design decision: separate static and dynamic data

I would use SQLite as the default database for structured operational data, but I would not force every dataset into the same update process.

Neighbourhood data: Build a new snapshot, validate it, and only then make it active. If validation fails, continue using the last successful snapshot.

Events: Store validated event records, track ingestion progress, and remove ended events according to the requirements.

Places: Regenerate and validate the place list separately from event ingestion.

Live parking: Cache responses briefly and refresh them when necessary.

The requirement to preserve the previous valid neighbourhood snapshot is especially important: a failed data refresh must not make the live application unusable. [cites: GoWhere_requirements_and_use_ca…]

When should the team migrate to PostgreSQL?

Consider PostgreSQL if GoWhere develops more demanding concurrent writes, requires a shared database across multiple application instances, or needs more sophisticated relational queries and operational controls.

SQLite's simplicity is valuable during early development, but it should not be treated as the automatic long-term answer for every deployment scenario.

One additional consideration is geographic querying. If the project eventually needs extensive spatial queries directly in the database, the team could evaluate PostgreSQL with PostGIS. However, I would not add that complexity unless the workload demonstrates a need for it.

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6. Data processing, maps, and external APIs

GoWhere's external integrations are central to its functionality, so I would prioritise reliability and clear interfaces over adding more infrastructure.

Recommended libraries and services

Requirement	Recommendation	Implementation approach
Neighbourhood calculations	pandas and NumPy	Process structured data and implement scoring calculations
Maps and routing	OneMap API	Resolve Singapore locations and retrieve routes
Parking availability	LTA DataMall	Retrieve current carpark availability
Event collection	Telegram integration	Fetch posts from configured public channels
Event extraction	An LLM API through `LlmAdapter`	Convert post content into structured event records
Place ratings	Google Places API	Retrieve ratings on demand within the configured daily cap
HTTP communication	Python `requests`	Centralise request handling, timeouts, and response parsing

The documents identify OneMap, DataMall, Telegram, Google Places, and the LLM service as external dependencies. They also specify that each external service should be accessed through its own adapter with a documented fallback. [cites: GoWhere_requirements_and_use_ca…]

The most important architectural rule: use adapters

Do not scatter direct API calls throughout the application.

Instead, make each adapter responsible for communicating with one external service.

For example:

```
class OneMapAdapter:
    def search(self, query):
        """Resolve a location using OneMap."""
        ...

    def route(self, start, end, route_type):
        """Retrieve route information."""
        ...
```

The implementation above is illustrative, not a complete working API client.

The calling service should depend on the adapter's interface rather than on the details of HTTP requests, authentication, and response formats.

This allows the team to:

- Replace an API provider without rewriting business logic.
- Test the application without making real API calls.
- Handle timeouts and failures consistently.
- Apply caching and rate limits in one place.
- Mock unavailable services during development.

This design is already reflected in the class diagram's boundary adapters and shared location and routing services. [cites: class_diagram_mermaid, class_diagram_mermaid]

External-service reliability

Each adapter should have:

1. A finite request timeout.
2. Appropriate handling for HTTP errors and malformed responses.
3. Validation of returned data.
4. A documented fallback or graceful degradation strategy.
5. Logging that records useful operational information without exposing secrets or private user inputs.
6. Unit tests using mocked responses.

For example, if OneMap cannot calculate a route, the event search should still display the event with a clearly labelled straight-line distance estimate where the requirements permit it.

If Google Places ratings are unavailable, food results should still be displayed without ratings, and rating-based sorting should be disabled.

These are specified behaviours, not merely optional enhancements. [cites: GoWhere_requirements_and_use_ca…, GoWhere_requirements_and_use_ca…]

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7. Testing and code quality

For an entry-level team, I would prioritise a strong automated testing workflow over adding advanced infrastructure.

Recommended testing stack

- **pytest:** Main test framework.
- **pytest-cov:** Coverage measurement.
- **unittest.mock:** Mock external APIs and isolate components.
- **Ruff:** Python linting and code-quality checks.
- **Black:** Optional code formatter if the team prefers it.
- **GitHub Actions:** Automatically run tests when code is pushed or a pull request is opened.

Three levels of testing

Level 1: Unit tests

Test individual components in isolation.

Examples:

- Does the scoring engine calculate weighted means correctly?
- Does the housing-affordability scorer enforce the minimum transaction threshold?
- Does the opening-hours parser handle a closed venue correctly?
- Does the search logic exclude events beyond the configured travel-time limit?

The scoring engine is particularly suitable for unit testing because the requirements specify exact calculation formulas and rounding behaviour. [cites: GoWhere_requirements_and_use_ca…]

Level 2: Integration tests

Test how components work together.

Examples:

- Can the event repository store and retrieve a valid event?
- Does the search service handle a mocked OneMap response?
- Does a failed data refresh leave the existing snapshot active?
- Does event ingestion preserve unprocessed posts when a dependent service fails?

Level 3: End-to-end tests

Test complete user workflows through the browser.

Examples:

- Compare neighbourhoods and view the recommendation.
- Search for an upcoming event and open its route.
- Search for food establishments and filter the results.

A browser automation tool such as Playwright can be added once the main workflows are stable.

What should be tested first?

I would prioritise:

1. Scoring calculations and ranking.
2. Input validation.
3. External API failure handling.
4. Data snapshot replacement.
5. Session isolation and privacy.
6. Search sorting and filtering.
7. End-to-end user workflows.

The requirements explicitly cover concurrent requests, external-service failures, data freshness, and session-only handling of location and other user inputs. These need dedicated tests rather than relying on a successful manual demonstration. [cites: GoWhere_requirements_and_use_ca…]

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8. Git, GitHub, and team collaboration

For a beginner team, I recommend using GitHub with a lightweight pull-request workflow.

Suggested workflow

1. Keep the main branch deployable.
2. Create a separate branch for each feature or bug fix.

3. Open a pull request before merging changes.
4. Require automated tests to pass.
5. Ask at least one teammate to review changes where team size permits.
6. Use GitHub Issues to track tasks and defects.
7. Document architectural decisions in the repository.

A reasonable branch naming convention is:

```
feature/neighbourhood-scoring
feature/event-search
feature/food-search
fix/onemap-timeout
test/scoring-engine
```

Why this matters for GoWhere

The three features share routing, location services, and some data sources. Parallel development without clear interfaces could cause duplicated logic and merge conflicts.

I would assign ownership of individual modules while requiring the team to agree on shared interfaces early.

For example, the team should agree on the interface for route calculations before separate developers implement the event-search and food-search workflows.

That is more important than adopting a complicated branching strategy.

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9. Deployment and scheduled tasks

Recommendation: start with a simple deployment

Use a managed Python hosting service or a small Linux server that supports the chosen web framework and persistent storage.

For a student project, avoid beginning with Kubernetes, multiple microservices, or a complex cloud architecture.

However, the deployment must support more than just serving web requests.

GoWhere needs:

- A persistent location for its database and validated data snapshots.
- Secure storage for API credentials.
- A way to run event ingestion periodically.
- A way to perform data refreshes and place-list builds.
- Logging and basic monitoring.
- A recovery procedure for failed updates.

The requirements specify that event ingestion runs every six hours when scheduling is enabled, while neighbourhood snapshots must be refreshed at least every three months. [cites:

GoWhere_requirements_and_use_ca…]

A practical setup would be:

Task	Recommended mechanism
Serve web requests	Flask application process
Ingest events	Scheduled job every six hours
Remove ended events	Daily scheduled job
Refresh neighbourhood snapshot	Maintainer-triggered job at least every three months
Rebuild the place list	Maintainer-triggered job
Run automated tests	GitHub Actions
Store secrets	Hosting-platform secret manager or environment variables
Monitor failures	Application logs and basic health checks

Important deployment caveat: Do not rely on an in-memory session store if the application will run across multiple workers or instances. The class diagram describes per-session routing and search state held in memory. For a multi-worker deployment, use a shared session store or explicitly design a reliable session-affinity strategy. A shared cache such as Redis may be appropriate if the deployment needs it, but it is unnecessary for the initial single-process setup.

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10. Security and privacy

GoWhere's security requirements should influence the architecture from the start.

The application must not require account creation or collect personally identifiable information. User location, postal codes, comparison weights, and factor options must remain session-scoped and must not be written to URLs, cookies, logs, or disk. API credentials must remain server-side. [cites: GoWhere_requirements_and_use_ca…]

I recommend the following controls:

Risk	Recommended control
API credentials accidentally committed to Git	Environment variables, secret management, and a `.gitignore` file
Credentials exposed in browser requests	Make external API calls from the backend wherever appropriate
SQL injection	Parameterised SQL queries
Cross-site scripting	Jinja's contextual autoescaping and validation of untrusted input
Malicious instructions in event posts	Treat post content strictly as data; validate all extracted event fields
Accidental storage of private search inputs	Explicitly restrict what may be persisted and logged
Excessive API calls	Rate limiting, request caching, and configured daily caps
Corrupted data refresh	Validate new snapshots before activation
External service outage	Timeouts, error handling, and graceful fallbacks

One subtle point: the requirement says session-only inputs must not be written to cookies or disk. Therefore, the team should not assume that conventional cookie-based session storage is acceptable. Choose a session design that satisfies the actual privacy constraints.

Similarly, an environment variable is suitable for keeping a secret out of source control, but a managed secrets service may offer stronger operational controls for a production deployment.

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11. What I would deliberately avoid

A beginner team can lose considerable time maintaining infrastructure that contributes little to the product's core functionality.

I would initially avoid the following:

Technology or approach	Why I would avoid it initially	When to reconsider
React or Next.js	Adds frontend framework and build complexity	Rich client-side interactions become a major requirement
Microservices	Introduces service communication, deployment, and monitoring overhead	Independent scaling or deployment becomes necessary
Kubernetes	Too much operational overhead for a small application	There is a demonstrated need for container orchestration
PostgreSQL plus PostGIS	More setup and maintenance than necessary for an initial SQLite deployment	Concurrent writes or advanced spatial queries justify the migration
Redis	Another service to deploy and maintain	Shared caching or session state becomes necessary
Celery and a message broker	More infrastructure than simple scheduled jobs initially require	Background workloads need queues, retries, and distributed workers
A full ORM migration	Could introduce unnecessary changes if the current data-access layer is already adequate	The team needs more sophisticated schema management or relational modelling

These are not inherently bad technologies. The recommendation is to introduce them in response to actual requirements rather than because they are common in larger production systems.

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12. Implementation roadmap for the team

I would introduce the stack in the following order.

Phase 1 — Establish a working application

Technologies: Python, Flask, HTML, CSS, Jinja, Git, and GitHub.

Tasks:

- Set up the repository and development environment.
- Create the Flask application and shared page layout.

- Implement navigation between the three features.
- Establish naming conventions and module responsibilities.
- Configure basic logging and error handling.

Deliverable: A working web application skeleton.

Phase 2 — Implement the core business logic

Technologies: pandas, NumPy, SQLite, pytest.

Tasks:

- Implement the neighbourhood scoring engine.
- Implement event and food search logic.
- Define the data-access interfaces.
- Add input validation and deterministic sorting.
- Write unit tests for the calculation and filtering rules.

Deliverable: The core application logic works independently of the external services.

Phase 3 — Integrate external services

Technologies: `requests`, OneMap API, LTA DataMall, Telegram, Google Places, and an LLM API.

Tasks:

- Implement one adapter per external service.
- Configure credentials securely.
- Add timeouts, response validation, and fallback behaviour.
- Mock external APIs in tests.
- Implement route and location caching where appropriate.

Deliverable: All three features can use their required external data sources without depending directly on API implementation details.

Phase 4 — Complete the user experience

Technologies: JavaScript, Chart.js, CSS.

Tasks:

- Implement interactive search forms.
- Display neighbourhood scores and recommendations.
- Add maps, route details, and progress indicators.
- Implement filtering, sorting, and pagination.
- Test responsive layouts on desktop and mobile browsers.

Deliverable: Users can complete the main workflows from start to finish.

Phase 5 — Reliability and deployment

Technologies: pytest-cov, Ruff, GitHub Actions, scheduled jobs, and a suitable Python hosting service.

Tasks:

- Add integration and end-to-end tests.
- Test external-service failures and concurrent requests.
- Verify data-refresh recovery.
- Configure scheduled ingestion and maintenance tasks.
- Deploy the application and verify its privacy controls.

Deliverable: A maintainable deployment that meets the most important functional and non-functional requirements.

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13. Final recommendation

If I were advising an entry-level team building GoWhere, I would settle on the following baseline:

Core stack

- Backend: Python + Flask
- Frontend: HTML + CSS + vanilla JavaScript + Jinja
- Data processing: pandas + NumPy
- Database: SQLite
- Visualisation: Chart.js
- Maps and routing: OneMap API
- External API calls: `requests`
- Testing: pytest
- Code quality: Ruff
- Version control: Git + GitHub
- CI: GitHub Actions
- Deployment: A managed Python host or small Linux server
- Scheduling: Cron or the hosting platform's scheduled-job facility

Three decisions matter more than adding more technologies:

1. **Maintain modularity.** Keep business logic separate from web routes, data access, and external-service adapters.
2. **Make failure handling a first-class requirement.** A failed API request or data refresh should not unnecessarily disable unrelated features.
3. **Automate testing and protect user data.** The scoring formulas, data-refresh process, external-service fallbacks, and session privacy constraints should all be verified systematically.

The proposed stack is deliberately conservative. It is capable of supporting GoWhere's three features while giving a beginner team a manageable codebase and a clear path to more sophisticated infrastructure if the project grows.

> The priority should be to deliver a reliable, testable application—not to maximise the number of technologies in the stack.

ChatGPT's file-citation markers are shown as [cites: …]: `GoWhere_requirements_and_use_ca…` is the requirements document, `class_diagram_mermaid` the class diagram.

Answers to 3.3.2

How similar are the recommendations? Mostly the same. The core is unchanged: Python, Flask, HTML/CSS/JavaScript, Chart.js, pytest and Git. Four things changed:

- storage moved from a JSON/CSV snapshot to SQLite
- GeoPandas was dropped
- maps moved from Leaflet to OneMap
- new advice was added: one adapter per external service, scheduled jobs, CI and security controls

(1) What caused the differences? Our prompt and preferences were the same, so the change comes from the inputs:

- **From the class diagram,** which Lab 2 did not have:
  - its adapter classes led to the "one adapter per service" rule
  - its three stores (`Snapshot`, `EventStore`, `PlaceStore`) led to SQLite in three stores
  - its in-memory `SessionStore` led to the single-process caveat
- **From the requirements:** the new features (Telegram, LLM, Google Places) added those services to the stack.

(2) Not applicable. We have not changed language or framework since Lab 2.

What we adopt, and what we don't:

- **Adopted:** Python, Flask, Jinja, vanilla JavaScript, Chart.js, SQLite, `requests`, pytest, one adapter per external service, and a single process with in-memory sessions.
- **Not adopted:**
  - *pandas/NumPy:* our scoring is medians and percentiles over about 10,800 HDB blocks, which plain Python handles, so it is one less dependency.
  - *OneMap for maps:* OneMap supplies location and routing data, but drawing a map needs a JavaScript library, so we use Leaflet and MapLibre.
  - *Serving with Flask itself:* Flask's built-in server is for development only, so we use Waitress.
  - *Ruff and GitHub Actions:* not adopted yet.

Part 2 — AI-generated minimal skeleton (3.3.3)

Uploaded: the class diagram only. Fresh chat.

Prompt:

> The attached file is the design class diagram of a web application, written in Mermaid.
>
> Generate a minimal skeleton template for this project in Python 3, using Flask.

> Rules for the files:
> 1. Create a folder hierarchy containing all the files you think the project needs (source, templates, tests, configuration).
> 2. Every file must be empty except for TODO comments describing what will go in it.
> 3. Do not write any code at all. No imports, no class or function definitions, no method signatures, no variables, no configuration values, no HTML tags. Only comments.
>
> Package the whole folder hierarchy as a single .zip file I can download.
>
> After the zip, show the folder tree, and give one sentence per folder explaining why it exists.

AI response:

I've created the **GoWhere Flask project skeleton** based on your Mermaid design class diagram. The structure follows the diagram's three main architectural layers: Boundary, Control, and Entity. [cites: class_diagram_mermaid]

1. Download the ZIP file

gowhere_flask_skeleton.zip📦 Download the GoWhere Flask Skeleton (sandbox:/mnt/data/gowhere_flask_skeleton.zip)

The ZIP contains 74 files, organized into a Python 3 and Flask project structure.

What I've included

- **Boundary layer:** Web routes and adapters for OneMap, DataMall, Google Places, Telegram, LLMs, data.gov.sg, and Overpass.
- **Control layer:** Search services, scoring engine and strategies, routing, location services, parking, and event ingestion.
- **Entity layer:** Events, places, neighbourhood snapshots, stores, opening hours, and session state.
- **Templates:** Placeholder pages for Live, Lepak, and Eat workflows, plus error pages.
- **Tests:** Placeholder test files for routes, search, scoring, adapters, routing, and stores.
- **Configuration:** Placeholder files for project dependencies, settings, environment variables, and the application entry point.

Your constraints are preserved: every file contains only TODO comments. There are no imports, class or function definitions, method signatures, variables, configuration values, or HTML tags.

---

2. Folder tree

```
gowhere_flask_skeleton/
├── README.md
├── pyproject.toml
├── requirements.txt
├── .env.example
├── .gitignore
├── run.py
│
├── config/
│   ├── __init__.py
│   └── settings.py
│
├── app/
│   ├── __init__.py
│   │
│   ├── boundary/
│   │   ├── __init__.py
│   │   ├── web_routes.py
│   │   ├── onemap_adapter.py
│   │   ├── token_source.py
│   │   ├── datamall_adapter.py
│   │   ├── llm_adapter.py
│   │   ├── telegram_post_source.py
│   │   ├── google_places_adapter.py
│   │   ├── daily_cap.py
│   │   ├── data_gov_adapter.py
│   │   └── overpass_adapter.py
│   │
│   ├── control/
│   │   ├── __init__.py
│   │   ├── lepak_search.py
│   │   ├── eat_search.py
│   │   ├── eat_ratings.py
│   │   ├── eat_hours.py
│   │   ├── scoring_engine.py
│   │   ├── scoring_strategy.py
│   │   ├── public_transport_scorer.py
│   │   ├── housing_affordability_scorer.py
│   │   ├── amenities_scorer.py
│   │   ├── greenery_scorer.py
│   │   ├── healthcare_scorer.py
│   │   ├── commute_scorer.py
│   │   ├── commute_router.py
│   │   ├── location_service.py
│   │   ├── event_router.py
│   │   ├── parking_service.py
│   │   └── ingestor.py
│   │
│   └── entity/
│       ├── __init__.py
│       ├── snapshot.py
│       ├── neighbourhood.py
│       ├── event_store.py
│       ├── event.py
│       ├── place_store.py
│       ├── place.py
│       ├── hours.py
│       ├── factor_choice.py
│       ├── session_store.py
│       └── session_state.py
│
├── templates/
│   ├── base.html
│   ├── live/
│   │   ├── submit.html
│   │   └── results.html
│   ├── lepak/
│   │   ├── submit.html
│   │   ├── results.html
```

```
│   │   └── event.html
│   ├── eat/
│   │   ├── submit.html
│   │   ├── results.html
│   │   ├── area.html
│   │   └── place.html
│   └── errors/
│       ├── 404.html
│       └── 500.html
│
├── static/
│   ├── css/
│   │   └── TODO.md
│   ├── js/
│   │   └── TODO.md
│   └── images/
│       └── TODO.md
│
└── tests/
    ├── __init__.py
    ├── conftest.py
    ├── test_web_routes.py
    ├── test_lepak_search.py
    ├── test_eat_search.py
    ├── test_eat_hours.py
    ├── test_scoring_engine.py
    ├── test_routing_services.py
    ├── test_stores.py
    └── test_adapters.py
```

---

3. Why each folder exists

Folder	Purpose
`config/`	Holds application configuration documentation and the planned settings structure, keeping configuration concerns separate from business logic.
`app/`	Contains the main application package and the components that implement the web application's architecture.
`app/boundary/`	Isolates the web interface and external-system integrations so the rest of the application does not need to handle API communication directly.
`app/control/`	Contains the application's workflows, business logic, search operations, scoring strategies, and routing services.
`app/entity/`	Organizes domain records, data-access components, stores, and session-state representations.
`templates/`	Holds the HTML template files that Flask will eventually render for the application's pages.
`templates/live/`	Groups the pages associated with the Live workflow.
`templates/lepak/`	Groups the event-search, event-results, and event-detail pages.
`templates/eat/`	Groups the food-search, browsing, results, and place-detail pages.
`templates/errors/`	Reserves templates for common HTTP error pages, such as 404 and 500.
`static/`	Reserves space for frontend assets served directly by Flask.
`static/css/`	Will hold the application's stylesheets.
`static/js/`	Will hold browser-side JavaScript.

`static/images/`	Will hold image assets used by the interface.
`tests/`	Separates automated tests from application code so individual components and their interactions can be verified.

Root-level files

These files are not folders, but they have distinct responsibilities:

File	Purpose
`README.md`	Documents the project's purpose, architecture, setup, and development workflow.
`pyproject.toml`	Reserves space for Python project metadata and development-tool configuration.
`requirements.txt`	Documents where runtime dependencies, including Flask, will eventually be listed.
`.env.example`	Documents the environment variables the application will eventually require without storing real secrets.
`.gitignore`	Documents which local files and generated artifacts should be excluded from version control.
`run.py`	Reserves the entry point for starting the application during development.
`app/__init__.py`	Documents the intended Flask application setup and application-factory responsibilities.
`tests/conftest.py`	Reserves space for shared pytest fixtures and test setup.

---

4. Architectural notes

I made three deliberate decisions when translating your diagram into a project structure.

1. Preserve the three-layer architecture.

The Boundary, Control, and Entity packages reflect the diagram's organization rather than placing all classes into a single application module. This should make the project easier to navigate as it grows.

2. Give the scoring strategies separate files.

The six scoring strategies each have their own placeholder file, alongside the shared strategy and scoring-engine files. This mirrors the Strategy pattern shown in the diagram and keeps individual scoring concerns separate.

3. Keep external integrations at the boundary.

Adapters for OneMap, DataMall, Google Places, LLMs, and other external services live under `app/boundary/`. Their consumers and business workflows are organized separately under `app/control/`.

One important limitation: this is a structural skeleton, not an executable Flask application yet. Since you requested comment-only files, Flask cannot run until the relevant TODOs are implemented.

> **Recommended next step:** implement the application factory and basic routing first, then add the Entity layer, Control services, and external adapters incrementally. This will let you establish a working Flask application before implementing the more complex business logic.

ChatGPT's file-citation markers are shown as [cites: …]: `GoWhere_requirements_and_use_ca…` is the requirements document, `class_diagram_mermaid` the class diagram.

The output is submitted as `lab3/ai_report/ai_generated_skeleton.zip`, separate from our application skeleton in `gowhere/`.

Answers to 3.3.4

To what extent do we agree with its choices?

- **We agree with the content.** The folders follow the diagram's boundary / control / entity layers, each scoring strategy gets its own file, and external services sit in the boundary layer.
- **We do not adopt its folder layout.** Our `gowhere/` groups code by feature instead (`scoring/`, `lepak/`, `eat/`, `adapters/`, …), so that each feature's changes stay in one folder.

Has it missed anything obvious?

- **No home page template.**
- **Nothing builds the data snapshot or the place list.** The data.gov.sg and Overpass adapters exist, but nothing calls them. Our class diagram did not show those build classes either, so the AI could not have known about them. We have since added them.
- **No `data/` folder.**
- **Only a development `run.py`.** There is no production server and no ingestion schedule.
- **Each TODO is a single line,** so the operations listed in the class diagram are lost.

Did it add anything beyond what was requested? No code: we checked that all 74 files contain only a comment. Small extras:

- placeholder `TODO.md` files in the `static/` folders
- a `pyproject.toml` that duplicates `requirements.txt`
- an unrequested "Architectural notes" section in its reply